

L^AT_EX Exam Extras — Part A

Code, Links, References, Footnotes, Lists, Bibliography
Exercises and Solutions

Reproduce Part I — check Part II afterwards

July 21, 2026

Abstract

This is Part A of a two-file set. It covers the elements of an exam paper that are individually small but collectively decisive: code listings, hyperlinks, every flavour of cross-reference, footnotes in awkward places, list customisation, the bibliography and page furniture. Part B, in the companion file `exam_extras_B.tex`, is a catalogue of features you may never have met, so that an unfamiliar element in a question paper becomes a lookup rather than a guess.

Rules of engagement. Print Part I only. Start from a blank file with your own preamble. Time each attempt. An element “passes” only if it *compiles* and matches the target.

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PART I — Target Elements

1 Reproduce each of the following

1.1 Exercise 1 — Code listing basics

Difficulty: ★★ **Target time:** 6 minutes

Tests: `lstlisting` with caption and label, line numbers, a frame, `lstinline`, cross-reference to a listing

Listing 1 implements the forward Euler method. The core update is the single expression $y += h * f(t, y)$, applied once per step.

```

1 def euler(f, t0, y0, h, n):
2     """Integrate y' = f(t, y) with n steps of size h."""
3     t, y = t0, y0
4     history = [(t, y)]
5
6     for _ in range(n):
7         y += h * f(t, y)      # forward step
8         t += h
9         history.append((t, y))
10
11    return history

```

Listing 1: Forward Euler integration.

The listing above is numbered automatically and can be referred to as Listing 1.

1.2 Exercise 2 — Several languages, several styles

Difficulty: ★★★ **Target time:** 8 minutes

Tests: per-listing language switching, a second named style, `escapeinside` for maths inside code

```

1 double euler(double (*f)(double, double),
2             double t, double y, double h, int n)
3 {
4     for (int i = 0; i < n; i++) {
5         y += h * f(t, y);    /* forward step */
6         t += h;
7     }
8     return y;
9 }

```

Listing 2: The same method in C.

Mathematics can be escaped into a listing so that a formula renders properly:

```

1 tau = R * C          τ = RC
2 v = V0 * (1 - exp(-t / tau))    v_C(t) = V_0(1 - e^{-t/τ})

```

Listing 3: Maths inside a listing.

1.3 Exercise 3 — The verbatim family

Difficulty: ★★ **Target time:** 5 minutes

Tests: `verbatim` environment, `verb` with different delimiters, showing L^AT_EX source literally

Reproduce this block exactly, including the backslashes:

```
\begin{figure}[htbp]
  \centering
  \includegraphics[width=0.5\textwidth]{plot.png}
  \caption{A caption.}
  \label{fig:plot}
\end{figure}
```

Inline, the command `\includegraphics` takes an optional argument, and `\textwidth` is a length. The starred form shows spaces explicitly: `two\llspaces`.

1.4 Exercise 4 — Hyperlinks

Difficulty: ★★ **Target time:** 6 minutes

Tests: `href`, `url`, `mailto`, internal links, `hypersetup` colours, PDF metadata

External link with custom text: [the Comprehensive TeX Archive Network](#). A bare address printed in monospace: `https://www.ctan.org/pkg/listings`. An email link: `someone@example.edu`. An internal link to a labelled element with text of your own choosing: [jump to the Euler listing](#).

The colours were set once in the preamble with `\hypersetup{colorlinks=true, linkcolor=RoyalBlue, ...}`, and this document's PDF title and author metadata were set in the same call.

1.5 Exercise 5 — The cross-reference gauntlet

Difficulty: ★★★ **Target time:** 8 minutes

Tests: `ref`, `eqref`, `pageref`, `autoref`, `nameref`, `cref`

Label the four elements below, then write the paragraph that follows using each referencing command exactly once.

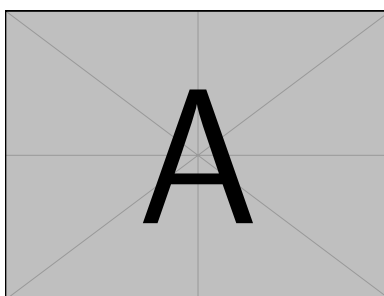


Figure 1: A figure to point at.

Table 1: A table to point at.

Quantity	Value
τ	5 ms
f_c	31.8 Hz

$$\tau = RC \tag{1}$$

compile twice

Now the paragraph. Plain number: Figure 1. Equation with parentheses: (1). Page number: Table 1 appears on page 4. Automatic type name: Figure 1. Section title by name: Exercise 5 — The cross-reference gauntlet. Clever reference, which supplies the word and handles plurals: fig. 1 and table 1 and capitalised at the start of a sentence: Equation (1) defines the time constant.

1.6 Exercise 6 — Footnotes in easy and awkward places

Difficulty: ★★★ **Target time:** 8 minutes

Tests: ordinary footnote, footnote inside a float, footnote in a caption, symbol footnotes, table notes

An ordinary footnote in body text.¹

A footnote from inside a table, which needs the two-part form:

Table 2: Measured values with a footnote marker.

Trial	Result
A	1.24
B	1.31 ²

A table with proper notes underneath, which is usually cleaner than a footnote:

Table 3: Using `threeparttable` for notes.

Material	E_g (eV)	Mobility ^a
Silicon	1.12	1350
Germanium	0.67	3900 ^b

^a Electron mobility at 300 K.

^b Value varies strongly with doping.

1.7 Exercise 7 — List customisation

Difficulty: ★★★ **Target time:** 9 minutes

Tests: three-level nesting with distinct markers, `description`, resumed numbering, an inline list, a two-column list

(I) Analysis methods

- † Time-domain solution of the differential equation.
- † Transform-domain solution.
 - ▷ Laplace transform for transients.
 - ▷ Fourier transform for steady state.
- ⊕ A one-off marker for a special case.

(II) Numerical methods

- † Explicit schemes.

¹The common case: place `\footnote` immediately after the word, with no space.

²Trial B was repeated after recalibration.

† Implicit schemes.

A description list:

Transfer function

The ratio of output to input in the s -domain.

Time constant

The time to reach 63.2% of the final value.

Numbering interrupted and then resumed:

1. First step.
2. Second step.

An interrupting paragraph.

3. Third step, continuing the count.

An inline list, which runs inside the paragraph: (a) resistance, (b) capacitance, and (c) inductance are the three passive parameters.

A list set in two columns:

- | | |
|-----------|-----------|
| • Mercury | • Mars |
| • Venus | • Jupiter |
| • Earth | • Saturn |

1.8 Exercise 8 — Citations and bibliography

Difficulty: ★★ **Target time:** 6 minutes

Tests: thebibliography, entry styles, citing with a locator, multiple citations

Write a paragraph that cites a book [1], a specific page of it [1, p. 245], two sources at once [2, 3], and a web resource [4]. The bibliography itself appears at the end of this document, before Part III.

1.9 Exercise 9 — Document furniture

Difficulty: ★★★ **Target time:** 10 minutes

Tests: abstract, TOC/LOF/LOT, running headers, line spacing, a two-column stretch, numbering depth

This document already demonstrates most of the target: an `abstract`, a `\tableofcontents` followed by `\listoffigures` and `\listoftables`, and a `fancyhdr` header carrying a title on the left and the page number on the right, with a grey footer.

Reproduce a passage set at one-and-a-half spacing:

This paragraph is set with increased leading so that it is visibly looser than the surrounding text. Line spacing is a global or scoped setting, not a per-line one, which is why it is wrapped in an environment.

And a two-column stretch inside a one-column document:

The `multicol` package balances its columns automatically. It is the correct tool when a paper shows a short two-column passage inside an otherwise single-column article, because the `twocolumn` class option would apply to the entire document instead of this passage alone. Columns may be separated by a rule and the gap between them is adjustable.

1.10 Exercise 10 — Full simulation

Difficulty: ★★★★★ **Target time:** 15 minutes

Tests: a self-contained passage combining a listing, a highlighted term, units, a footnote, a coloured box, and three kinds of cross-reference

Worked example: extracting the time constant

Given the measurement in table 1, the time constant is $\tau = 5$ ms, so the cut-off frequency^a is

$$f_c = \frac{1}{2\pi\tau} = 31.8 \text{ Hz.}$$

The quantity $\tau = RC$ of eq. (1) is the only parameter needed; everything else follows. Listing 1 can integrate the response numerically if an analytic solution is unavailable.

^aDefined as the -3 dB point.

Terms may be struck out when superseded: the older estimate was ~~4.2 ms~~, now corrected. In a derivation, terms that vanish can be cancelled: $\frac{\cancel{Y_0}}{\cancel{Y_0}R} = \frac{1}{R}$. Small fractions inline read better as $1/2$ than as $\frac{1}{2}$.

PART II — Solutions and Marking Notes

2 Do not read until you have attempted Part I

Solution 1 — Code listing basics

```
% ---- preamble ----
\usepackage{listings}
\usepackage{xcolor}
\lstdefinestyle{py}{
  language=Python, basicstyle=\ttfamily\small,
  keywordstyle=\color{RoyalBlue}\bfseries,
  commentstyle=\color{OliveGreen}\itshape,
  stringstyle=\color{BrickRed},
  numbers=left, numberstyle=\tiny\color{gray}, numbersep=8pt,
  frame=single, breaklines=true, showstringspaces=false,
  tabsize=4, captionpos=b}

% ---- body ----
\Cref{lst:euler} implements the forward Euler method. The core update is
\lstinline[language=Python]!y += h * f(t, y)!, applied once per step.

\begin{lstlisting}[style=py, caption={Forward Euler integration.},
                  label={lst:euler}]
def euler(f, t0, y0, h, n):
    """Integrate y' = f(t, y) with n steps of size h."""
    t, y = t0, y0
    history = [(t, y)]

    for _ in range(n):
        y += h * f(t, y)      # forward step
        t += h
        history.append((t, y))

    return history
\end{lstlisting}
```

Marking notes. A listing is *verbatim*: everything between `\begin{lstlisting}` and `\end{lstlisting}` is printed exactly, including backslashes, quotes and indentation. That is why you can paste code straight in without escaping anything.

Define the style **once** in the preamble with `\lstdefinestyle`, then apply it with `[style=py]`. Retyping a dozen key–value options at every listing is the slow way and invites typos.

The options that earn the marks: `language=` switches the keyword dictionary; `numbers=left` adds line numbers; `frame=single` draws the box (other values are `lines`, `shadowbox`, `trBL`); `caption=` produces the numbered “Listing 1:” line and `label=` makes it referenceable; `captionpos=b` puts the caption below; `breaklines=true` wraps long lines instead of overflowing the margin; `showstringspaces=false` removes the ugly visible-space markers inside strings.

`\lstinline` is the inline equivalent of `\verb`. Both take *any* delimiter that does not occur in the code — here `!...!` was used because the code contains no exclamation mark. From an external file, use `\lstinputlisting[style=py]{euler.py}`, which is what a paper means when it says “using an external Python file”.

Solution 2 — Languages, styles, escaped maths

```
\lstdefinestyle{c}{
```



```

language=C, basicstyle=\ttfamily\small,
keywordstyle=\color{Purple}\bfseries,
commentstyle=\color{gray}\itshape, stringstyle=\color{BrickRed},
numbers=left, numberstyle=\tiny\color{gray},
frame=lines, breaklines=true, captionpos=b}

\begin{lstlisting}[style=c, caption={The same method in C.},
                  label={lst:eulerc}]
double euler(double (*f)(double, double),
             double t, double y, double h, int n)
{
    for (int i = 0; i < n; i++) {
        y += h * f(t, y); /* forward step */
        t += h;
    }
    return y;
}
\end{lstlisting}

\begin{lstlisting}[style=py, escapeinside={(*){(*)}},
                  caption={Maths inside a listing.}]
tau = R * C
      (*$\tau = RC$*)
\end{lstlisting}

```

Marking notes. `listings` knows several dozen languages out of the box, including C, C++, Java, Python, Matlab, R, SQL, bash, HTML, VHDL and [LaTeX]TeX. You can set the language globally with `\lstset{language=C}`, per style, or per listing.

`escapeinside={(*){(*)}}` declares a pair of markers that temporarily leave verbatim mode, so real L^AT_EX — usually maths — can be embedded in the code. Choose markers that cannot appear in the language; `(*` and `*)` are safe in C and Python but would be wrong in Pascal, where they are comment delimiters.

If a paper shows genuinely beautiful syntax colouring with a coloured background, that is often minted rather than `listings`. `minted` needs Python’s Pygments installed and compilation with `-shell-escape`; if it is unavailable, reproduce the appearance with `listings` plus `backgroundcolor=\color{black!5}` and move on.

Solution 3 — The verbatim family

```

\begin{verbatim}
\begin{figure}[htbp]
  \centering
  \includegraphics[width=0.5\textwidth]{plot.png}
  \caption{A caption.}
  \label{fig:plot}
\end{figure}
\end{verbatim}

```

Inline, `\verb|\includegraphics|` takes an optional argument, and `\verb+\textwidth+` is a length. The starred form shows spaces explicitly: `\verb*|two spaces|`.

Marking notes. Three tools, three jobs. The `verbatim` environment prints a block literally in monospace with no colouring or frame — it is the plainest way to show source. `\verb` does the same for a short inline fragment; the character immediately after it becomes the delimiter, so `\verb|...|`, `\verb+...+` and `\verb...!!` are all valid. The starred `\verb*` renders spaces as visible marks.

The trap that costs marks: `\verb` cannot be used inside another command’s argu-

ment. It will fail inside `\section{}`, `\caption{}`, `\footnote{}` or a table cell in some contexts. The workaround is `\texttt{\textbackslash includegraphics}`, spelling the backslash out as `\textbackslash`.

The `verbatim` package also provides a `comment` environment, which discards everything inside it — the clean way to disable a whole block during an exam without deleting it:

```
\begin{comment}
This entire block is ignored by the compiler.
\end{comment}
```

Solution 4 — Hyperlinks

```
\usepackage{hyperref}
\hypersetup{
  colorlinks = true,
  linkcolor  = RoyalBlue,    % internal: \ref, TOC, \hyperref
  citecolor  = teal,         % \cite
  urlcolor   = magenta,      % \href, \url
  pdftitle   = {Document title},
  pdfauthor  = {Your name},
  pdfsubject = {Subject},
  pdfkeywords= {keyword, keyword}
}

\href{https://www.ctan.org}{the Comprehensive TeX Archive Network}
\url{https://www.ctan.org/pkg/listings}
\href{mailto:someone@example.edu}{someone@example.edu}
\hyperref[lst:euler]{jump to the Euler listing}
```

Marking notes. `\href{destination}{visible text}` takes the URL first and the text second — reversing them is a common slip. `\url{...}` prints the address itself in monospace and, unlike `\href`, will break a long address across lines sensibly.

`\hyperref[label]{text}` is the internal cousin of `\href`: it links to a `\label` but lets you write your own wording instead of a bare number. That is how you make “see the results section” clickable.

Three appearance choices are worth knowing. `colorlinks=true` colours the text; the default instead draws coloured *boxes*, which look wrong in print. `hidelinks` removes all colouring and boxes while keeping the links live — ideal when a printed target shows plain black text that is nonetheless clickable. Setting all colours to `black` achieves the same look while leaving them technically coloured.

Two ordering rules: load `hyperref` **last** among ordinary packages, because it patches many internals; and load `cleveref` **after** `hyperref`, which is the one documented exception. Special characters in a URL — `%`, `#`, `_` — must be escaped inside `\href`, though `\url` handles them itself.

Solution 5 — The cross-reference gauntlet

```
Plain number: Figure~\ref{fig:target}.
Equation with parentheses: \eqref{eq:tau}.
Page number: Table~\ref{tab:target} appears on page~\pageref{tab:target}.
Automatic type name: \autoref{fig:target}.
Section title by name: \nameref{sec:refs}.
Clever reference: \cref{fig:target,tab:target}
and capitalised: \Cref{eq:tau} defines the time constant.
```

Marking notes. Six commands, six different outputs from the same label:

Command	Output	Supplies the word?
<code>\ref{k}</code>	1	no
<code>\eqref{k}</code>	(1)	no, adds parentheses
<code>\pageref{k}</code>	7	page number instead
<code>\autoref{k}</code>	Figure 1	yes (hyperref)
<code>\nameref{k}</code>	the section's title	the title itself
<code>\cref{k}</code>	fig. 1	yes (cleveref)

`\cref` is the most powerful: it supplies the type word, handles lists (`\cref{a,b,c}` gives “figs. 1, 2 and 3”), and collapses ranges. `\Cref` is its capitalised form for the start of a sentence. Rename the types with `\crefname{figure}{Fig.}{Figs.}` if the target uses abbreviations.

The universal rules: `\label` must come **after** `\caption`, every reference needs a **second compile** to resolve, and a ?? in the output means “run again”, not “your code is broken”. Prefix labels by type — `fig:`, `tab:`, `eq:`, `sec:`, `lst:` — because it prevents collisions and makes `\cref` behave predictably.

Solution 6 — Footnotes in awkward places

```
% ordinary
An ordinary footnote in body text.\footnote{The common case.}

% inside a float: two-part form
\begin{table}[H]
  \centering
  \caption{Measured values with a footnote marker.}
  \begin{tabular}{@{}lc@{}}
    \toprule Trial & Result \\ \midrule
    A & \num{1.24} \\
    B & \num{1.31}\footnotemark \\
    \bottomrule
  \end{tabular}
\end{table}
\footnotetext{Trial B was repeated after recalibration.}

% footnote in a section heading needs \protect
\section{A title\protect\footnote{Note text.}}

% symbol footnotes instead of numbers
\renewcommand{\thefootnote}{\fnsymbol{footnote}}

% proper table notes
\begin{threeparttable}
\caption{Using threeparttable for notes.}
\begin{tabular}{@{}lcc@{}}
\toprule
Material & $E_g$ & Mobility\tnote{a} \\ \midrule
Silicon & 1.12 & \num{1350} \\
\bottomrule
\end{tabular}
\begin{tablenotes}
\item[a] Electron mobility at \SI{300}{kelvin}.
\end{tablenotes}
\end{threeparttable}
```

Marking notes. A plain `\footnote` fails inside a float — the marker appears but the text is swallowed, because floats are typeset separately from the page they land on. The fix

is the two-part form: `\footnotemark` inside the float to print and advance the marker, then `\footnotetext{...}` immediately *after* the float to place the text.

In a *moving argument* such as a section title or a caption, protect it: `\protect\footnote{...}`, otherwise the footnote is re-executed when the title is written to the table of contents.

`\renewcommand{\thefootnote}{\fnsymbol{footnote}}` switches from 1, 2, 3 to *, †, ‡, §, ¶ — which is exactly what a paper is showing when its footnote markers are symbols rather than digits. Restore with `{\arabic{footnote}}`.

For notes attached to a specific table, `threeparttable` is cleaner and is what most published tables actually use: `\tnote{a}` in the cell, `\item[a]` in the `tablenotes` block, and the notes sit directly under the table at its own width rather than at the bottom of the page.

Solution 7 — List customisation

```
\usepackage{enumitem} % and for inline lists:
\usepackage[inline]{enumitem}

\begin{enumerate}[label=(\Roman*)]
  \item \textbf{Analysis methods}
    \begin{itemize}[label=$\dagger$]
      \item Time-domain solution.
      \item Transform-domain solution.
      \begin{itemize}[label=$\triangleright$]
        \item Laplace transform for transients.
        \item[$\oplus$] A one-off marker for a special case.
      \end{itemize}
    \end{itemize}
\end{enumerate}

\begin{description}[leftmargin=2cm, style=nextline]
  \item[Transfer function] The ratio of output to input.
\end{description}

\begin{enumerate}[label=\arabic*.] \item First. \item Second. \end{enumerate}
An interrupting paragraph.
\begin{enumerate}[label=\arabic*., resume] \item Third. \end{enumerate}

\begin{enumerate*}[label=(\alph*)]
  \item resistance, \item capacitance, and \item inductance
\end{enumerate*}

\begin{multicols}{2}
\begin{itemize}[noitemsep] \item Mercury \item Venus \end{itemize}
\end{multicols}
```

Marking notes. The counter commands inside `label=` are `\arabic*`, `\alph*`, `\Alph*`, `\roman*` and `\Roman*`; the star means “the current item’s number”. Anything else in the label is literal, so `label=(\Roman*)` gives (I), (II) and `label=Step \arabic*` gives “Step 1”.

Frequently useful keys: `resume` continues a previous list’s numbering across an interruption; `start=5` begins at five; `noitemsep` and `nosep` tighten vertical space; `leftmargin=` controls indentation; `itemsep=`, `topsep=`, `parsep=` tune it precisely; `widest=` aligns description terms; and `style=nextline` puts a description body on the following line.

Two discoveries most people miss. Loading with `\usepackage[inline]{enumitem}` enables `enumerate*`, `itemize*` and `description*`, which typeset a list *inside a paragraph* — that is how a paper shows “(a) x, (b) y and (c) z” running in the text. And `\item[$\oplus$]` overrides the marker for a single item without disturbing the rest.

To make a change global rather than per-list, use `\setlist[itemize,1]{label=$\bullet$}` in the preamble.

Solution 8 — Citations and bibliography

```
% ---- in the text ----
a book~\cite{kreyszig}, a specific page~\cite[p.~245]{kreyszig},
two at once~\cite{ogata,shannon}, a web resource~\cite{ctan}.

% ---- at the end of the document ----
\begin{thebibliography}{9}
  \bibitem{kreyszig} E.~Kreyszig, \textit{Advanced Engineering
    Mathematics}, 10th ed. Hoboken, NJ: John Wiley & Sons, 2011.
  \bibitem{ogata} K.~Ogata, \textit{Modern Control Engineering},
    5th ed. Upper Saddle River, NJ: Prentice Hall, 2010.
  \bibitem{shannon} C.~E.~Shannon, ‘‘A mathematical theory of
    communication,’’ \textit{Bell System Technical Journal}, vol.~27,
    no.~3, pp.~379--423, 1948.
  \bibitem{ctan} CTAN, ‘‘The Comprehensive \TeX{} Archive Network,’’
    2026. [Online]. Available: \url{https://www.ctan.org}
\end{thebibliography}
```

Marking notes. For a timed exam, `thebibliography` is almost always the right choice: it needs no external `.bib` file and no extra compilation passes. The mandatory argument `{9}` only reserves label width — use `{99}` if you have ten or more entries.

`\cite[p.~245]{key}` adds a locator, printing [1, p. 245]. `\cite{a,b}` cites several at once and is *not* the same as `\cite{a}\cite{b}`, which prints two separate brackets. Entries are numbered in the order the `\bibitems` appear, not the order they are cited.

Punctuation conventions that graders look at: book and journal *titles* in italic via `\textit{}`, article titles in ‘‘quotes’’ using two backticks and two apostrophes, an en-dash for page ranges (pp.~379--423), `&` escaped, and `~` after initials so E.~Kreyszig never splits across a line.

The alternative, for completeness, is a `.bib` database:

```
% refs.bib
@book{kreyszig,
  author   = {Kreyszig, Erwin},
  title    = {Advanced Engineering Mathematics},
  edition  = {10th}, publisher = {Wiley}, year = {2011}
}

% in the document
\bibliographystyle{ieeetr} % or plain, unsrt, abbrv, apalike
\bibliography{refs}

% compile: pdflatex -> bibtex -> pdflatex -> pdflatex
```

That four-pass cycle is why it is the wrong tool under time pressure, unless the paper explicitly demands it.

Solution 9 — Document furniture

```
% ---- abstract (article class, built in) ----
\begin{abstract}
Text of the abstract.
\end{abstract}

% ---- contents lists ----
\tableofcontents
\listoffigures
```

```

\listoftables
\setcounter{tocdepth}{2}      % how deep the TOC goes
\setcounter{secnumdepth}{3}   % how deep numbering goes
% add an unnumbered heading to the TOC:
\section*{Acknowledgements}
\addcontentsline{toc}{section}{Acknowledgements}

% ---- running headers and footers ----
\usepackage{fancyhdr}
\pagestyle{fancy}
\fancyhf{}                    % clear everything first
\fancyhead[L]{\small\itshape Short title}
\fancyhead[R]{\small\thepage}
\fancyfoot[C]{\small\textcolor{gray}{draft}}
\renewcommand{\headrulewidth}{0.4pt}

% ---- line spacing ----
\usepackage{setspace}
\onehalfspacing               % whole document
\begin{spacing}{1.5} ... \end{spacing} % one passage

% ---- a two-column stretch ----
\usepackage{multicol}
\begin{multicols}{2} ... \end{multicols}

```

Marking notes. `\fancyhf{}` clearing everything first is essential; without it the class’s default header survives and you get doubled content. Position arguments are [L], [C], [R], and in a two-sided document [LE,R0] means “left on even, right on odd”. `\thepage` is the page-number command.

`\setcounter{tocdepth}{2}` controls what the TOC *lists*; `\setcounter{secnumdepth}{3}` controls what gets *numbered*. Levels are: part = −1, chapter = 0, section = 1, subsection = 2, subsubsection = 3. A starred heading is unnumbered and skipped by the TOC, so add it back manually with `\addcontentsline` when a target shows an unnumbered heading that nevertheless appears in the contents.

`multicol` balances columns automatically and works mid-document, which the `twocolumn` class option cannot. Note that floats do not work inside `multicols` — use `\captionof` inside a `minipage` instead.

Solution 10 — Full simulation

```

\begin{tcolorbox}[colback=RoyalBlue!5, colframe=RoyalBlue!60,
  title=Worked example: extracting the time constant]
Given the measurement in \cref{tab:target}, the time constant is
 $\tau = \SI{5}{\milli\second}$ , so the cut-off frequency\footnote{Defined
as the  $\SI{-3}{\decibel}$  point.} is
 $f_c = \frac{1}{2\pi\tau} = \SI{31.8}{\hertz}$ . \]
The quantity \colorbox{yellow}{ $\tau = RC$ } of \cref{eq:tau} is the only
parameter needed. \Cref{lst:euler} can integrate the response numerically.
\end{tcolorbox}

the older estimate was \sout{4.2 ms}, now corrected.

$$\frac{\cancel{V_0}}{\cancel{V_0}R} = \frac{1}{R}$$

inline fractions read better as \nicefrac{1}{2}

```

Marking notes. Five small packages, each solving one visible problem. `tcolorbox` draws a titled coloured panel — `colback` is the fill, `colframe` the border, `title` the header strip. `siunitx` sets units upright with correct spacing: `\SI{5}{\milli\second}` gives 5 ms and `\num{1350}`

formats a bare number; this is far safer than typing `5 ms` by hand, which renders as italic variables. `soul` provides `\hl{}` highlighting; `ulem` provides `\sout{}` strike-out — load it as `\usepackage[normalem]{ulem}` or it silently redefines `\emph` to underline everything. `cancel` draws the diagonal strike used when terms vanish in a derivation, and `\cancelto{0}{x}` labels the limit. `nicefrac` sets small slanted inline fractions.

3 Drill Routine

1. **Round 1.** Exercises 1, 3, 4 untimed with Part II open. These are the three highest-frequency items: a listing, verbatim source, and links.
2. **Round 2.** Exercise 5 five times, until all six referencing commands are automatic. Cross-references are cheap marks and are almost always on the mark sheet.
3. **Round 3.** Exercises 6 and 7 in 20 minutes. Footnotes inside floats and custom list labels are the two that most often go wrong.
4. **Round 4.** Exercises 8 and 9 in 18 minutes, writing the bibliography entries from memory in the correct punctuation style.
5. **Round 5.** Exercise 10 cold in 15 minutes.
6. **Round 6.** Move to the companion file `exam_extras_B.tex` and read the Discovery Catalogue end to end. Then close it and list every package you can recall together with what it does. Anything you cannot recall, reread. You are not memorising syntax — only the *existence* of the tool and its name.

4 The Ten Errors That Cost the Most Marks

#	Mistake	Symptom and fix
1	<code>\verb</code> inside another command	Compilation error inside <code>\caption</code> , <code>\section</code> or <code>\footnote</code> . Use <code>\texttt{\textbackslash cr}</code> instead.
2	<code>\footnote</code> inside a float	Marker appears, text vanishes. Use <code>\footnotemark</code> plus <code>\footnotetext</code> after the float.
3	<code>hyperref</code> not loaded last	Broken links or clashes. Load it after everything except <code>cleveref</code> .
4	<code>cleveref</code> before <code>hyperref</code>	Package error. <code>cleveref</code> is the one that must come after.
5	<code>\label</code> before <code>\caption</code>	<code>\ref</code> gives the previous float's number. Caption first, always.
6	<code>ulem</code> without <code>normalem</code>	Every <code>\emph</code> in the document becomes underlined.
7	Unescaped The link truncates. Escape inside <code>\verb</code> , or use <code>\url</code> .	
8	Floats inside <code>multicols</code>	They silently disappear. Use <code>\captionof</code> inside a <code>minipage</code> .
9	<code>\fancyhf {}</code> omitted compile twice	Doubled headers. Clear before setting.

References

- [1] E. Kreyszig, *Advanced Engineering Mathematics*, 10th ed. Hoboken, NJ: John Wiley & Sons, 2011.
- [2] K. Ogata, *Modern Control Engineering*, 5th ed. Upper Saddle River, NJ: Prentice Hall, 2010.
- [3] C. E. Shannon, “A mathematical theory of communication,” *Bell System Technical Journal*, vol. 27, no. 3, pp. 379–423, 1948.
- [4] CTAN, “The Comprehensive T_EX Archive Network,” 2026. [Online]. Available: <https://www.ctan.org>

End of Part A. The Discovery Catalogue continues in `exam_extras_B.tex`.